ParkScreen — Screening Report

Multimodal decision-aid for Parkinson's disease signs — *not a diagnosis*.

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OVERALL RISK GRADE

Elevated likelihood of features consistent with PD

Fused score 0.89

Modality scores

Vocal	Facial
weight 100%	weight 0%
	
Speech channels: phonation 0.96, DDK 0.85.	*No smile video provided — Facial channel is N/A.*

Vocal channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Phonation 35% within modality	0.96	1.00	jitter 1.87%, HNR 18.2 dB, F0 SD 20.1 Hz
Articulation (DDK) 65% within modality	0.85	0.33	rate 4.74 syl/s, ISI CV 0.66, amp CV 0.40

Facial channel details

CHANNEL	SCORE	CONFIDENCE	NOTE
Smile task 0% within modality	N/A	—	all smile clips withheld by quality gate

Consistency

Channels agree

All active channels point in the same direction. Higher confidence in the fused score.

Clinical narrative

Risk Level

Elevated. The fused PD screening probability is **0.886** (weighted across the two available speech channels). This level of screening evidence is consistent with features frequently observed in Parkinson's disease and warrants specialist evaluation, but it is not a diagnosis.

Phonation

The sustained-vowel analysis shows markedly elevated cycle-to-cycle perturbation: **jitter 1.87%** and **shimmer 10.10%**, both well above typical adult ranges (~0.3–0.5% and ~3–5% respectively). The **HNR of 18.16 dB** is reduced, consistent with an increased noise component in the voice signal. **F0 mean 151.8 Hz** with **F0 std 20.1 Hz** and **range 57.3 Hz** are within a plausible speaking range but combine with the perturbation picture to yield a channel probability of **0.955**, consistent with PD-typical dysphonia.

Articulation (DDK)

The PATAKA task shows a **syllable rate of 4.75 Hz**, below the typical adult benchmark of ≥6 Hz, which may suggest bradykinetic articulation. Timing regularity is notably degraded (**ISI CV 0.663**), and amplitude variability is also elevated (**amp CV 0.403**), while amplitude decrement is essentially flat (**–0.0002**), so fading across the utterance is not a prominent feature here. The channel probability of **0.848** is driven primarily by the slow, irregular rhythm.

Cross-Channel Consistency

Phonation (0.955) and DDK (0.848) **agree** — both speech channels point in the same direction, which strengthens the overall screening signal. The **facial channel was omitted** (no smile upload), so hypomimia could not be assessed; this reduces multimodal coverage, and any facial motor signs would not be captured by this report.

Recommended Next Steps

Given the Elevated fused probability and consistent speech-channel findings, **referral to a movement-disorder neurologist** is recommended for clinical evaluation. Submitting a **smile-task recording** would allow the facial channel to contribute and improve the completeness of screening. If clinically indicated, repeat speech recording under standardized conditions may also help track features over time.

Disclaimers

This report is a screening decision-aid, not a clinical diagnosis. Any risk indication requires evaluation by a qualified movement-disorder specialist.

The classifiers were trained on Parkinson's-disease subjects recorded 2–5 hours post-medication (ON state). Off-state PD may present with more perturbed features than the training distribution, so this system's calibrated probability does not extrapolate to OFF-state inputs.

Screening decision-aid, not a diagnosis. ParkScreen provides supporting evidence for clinical review only. It is not a substitute for evaluation by a qualified healthcare professional and must not be used for self-diagnosis.